



Indian Institute of Technology Hyderabad

Department of Electrical Engineering

Demonstration of Sampling Theorem and Signal Reconstruction

Electronics Devices and Circuits Laboratory

Name: Muthyala Koushik

Roll Number: EE24BTECH11044

Contents

1	Objective	2
2	Apparatus	2
3	Theory	2
3.1	Nyquist Sampling Theorem	2
3.2	Sampling Frequency (From Code)	2
3.3	Aliasing	2
4	Sample and Hold Circuit	3
4.1	Sample and Hold Code	3
5	Successive Approximation Register (SAR) ADC	3
5.1	Working Principle	3
5.2	Ring Counter	5
5.3	SAR Code	5
6	R-2R Ladder DAC	6
7	ADC Performance Parameters (Detailed Explanation)	7
7.1	Resolution	7
7.2	LSB Voltage	8
7.3	Conversion Time	8
7.4	Offset Error	8
7.5	Gain Error	8
7.6	Differential Non-Linearity (DNL)	8
7.7	Integral Non-Linearity (INL)	8
8	Observations	8
8.1	Calculated Values	9
8.2	LSB Voltage	9
8.3	Conversion Time	10
8.4	Offset Error	10
8.5	Gain Error	11
8.6	Differential Non-Linearity (DNL)	11
8.7	Integral Non-Linearity (INL)	12
9	Conclusion	12

1 Objective

To experimentally verify the Sampling Theorem by sampling an analog signal at different frequencies using a Sample-and-Hold circuit, converting it to digital form using a 4-bit SAR ADC, reconstructing it using an R-2R ladder DAC, and observing aliasing and faithful reconstruction.

2 Apparatus

- Arduino Uno
- Comparator
- Resistors (R and $2R$)
- Function Generator
- CRO
- Low-Pass Filter
- Breadboard and connecting wires

3 Theory

3.1 Nyquist Sampling Theorem

According to the Nyquist Sampling Theorem, a band-limited signal of maximum frequency f_m can be perfectly reconstructed if the sampling frequency satisfies:

$$f_s \geq 2f_m$$

3.2 Sampling Frequency (From Code)

From the Arduino program:

$$T_{high} = 9000 \mu s, \quad T_{low} = 1000 \mu s$$

$$T_s = T_{high} + T_{low} = 10000 \mu s = 10 ms$$

$$f_s = \frac{1}{T_s} = \boxed{100 Hz}$$

3.3 Aliasing

When the sampling frequency is less than twice the signal frequency, spectral overlap occurs, producing distortion known as aliasing.

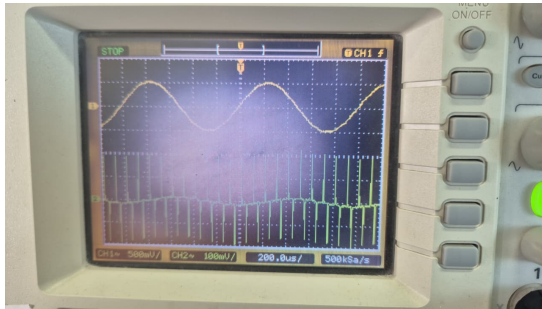
The aliased frequency is given by:

$$f_{alias} = |f_{signal} - nf_s|$$

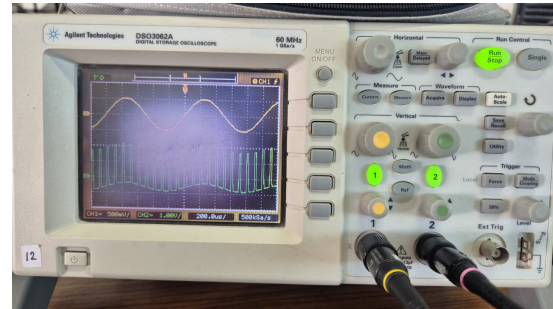
4 Sample and Hold Circuit

The Sample-and-Hold circuit captures the instantaneous value of the analog input signal and maintains it constant during the entire ADC conversion process. This is essential in SAR ADCs because the input must remain unchanged while successive bit decisions are made.

Sampling occurs on the falling edge of the sampling clock generated by the Arduino.



(a) Sampling Phase



(b) Hold Phase

4.1 Sample and Hold Code

```
if (slowClkState == LOW && prevSlowClk == HIGH) {  
    ring = 0b10000;  
    int adcVal = analogRead(vinPin);  
    VinHeld = (adcVal / 1023.0) * ADCref;  
}
```

The above code samples the analog input voltage and stores it as a constant value for the SAR conversion cycle.

5 Successive Approximation Register (SAR) ADC

5.1 Working Principle

The SAR ADC performs analog-to-digital conversion using a binary search algorithm. Each conversion cycle requires one comparison per bit, starting from the MSB and proceeding to the LSB.

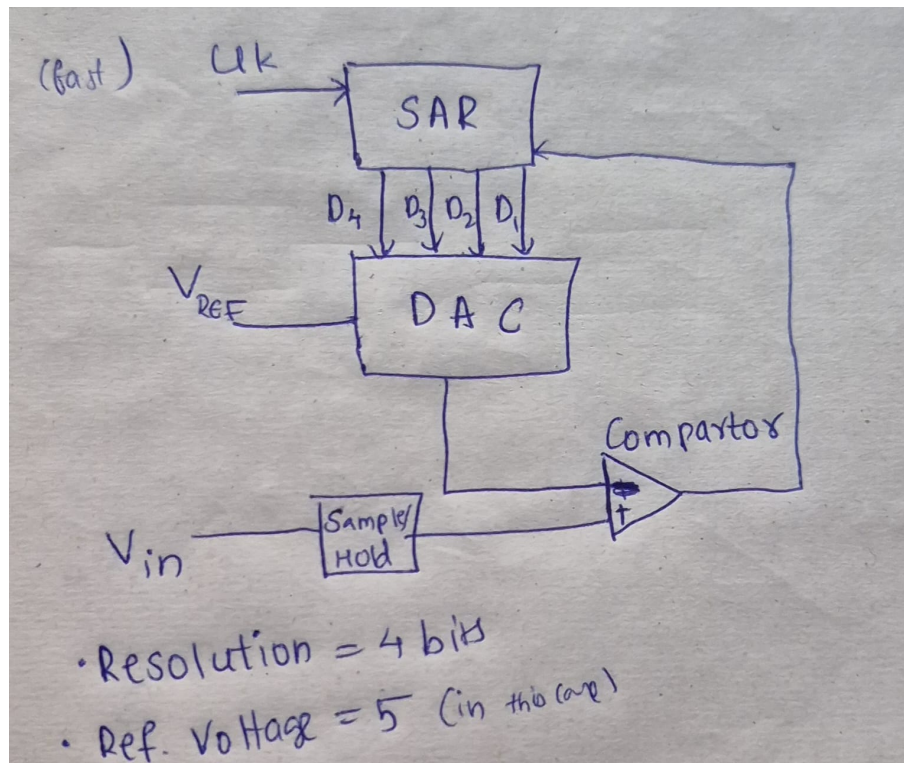


Figure 2: SAR Logic Block Diagram

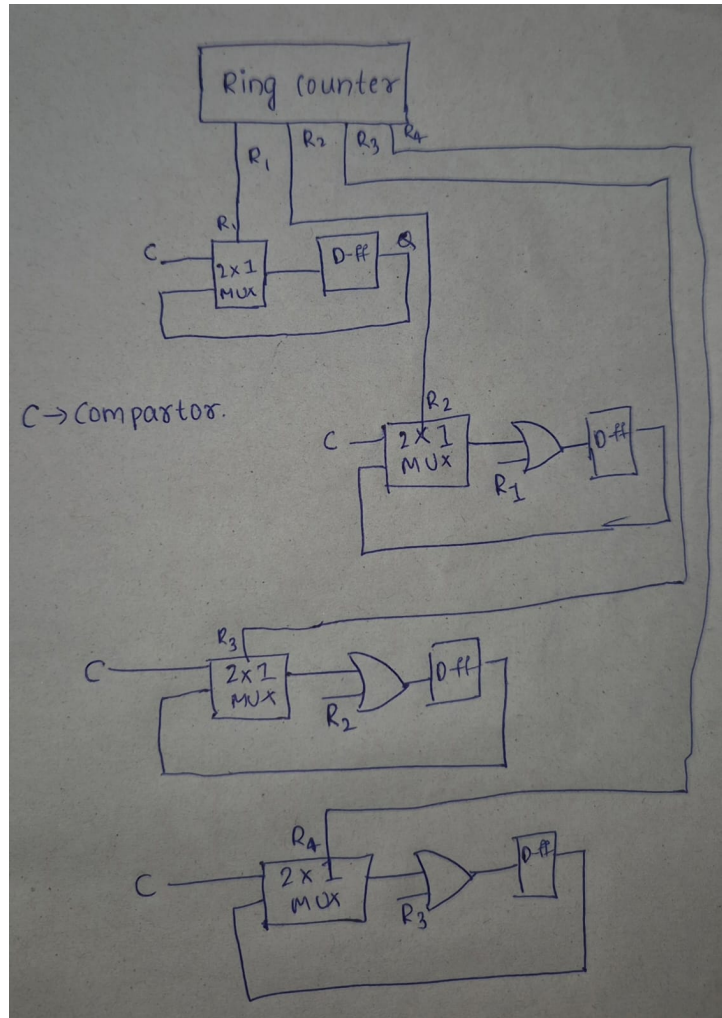


Figure 3: Ring Counter for Sequential Bit Selection

5.2 Ring Counter

The ring counter activates only one bit at a time, ensuring orderly MSB-to-LSB testing. After each comparator decision, the active bit shifts to the next lower significance.

5.3 SAR Code

```
float dacVoltage = (sarValue / 15.0) * Vref;
digitalWrite(comparatorPin, dacVoltage <= VinHeld ? HIGH : LOW);

if (ring != 0)
    ring >>= 1;
```

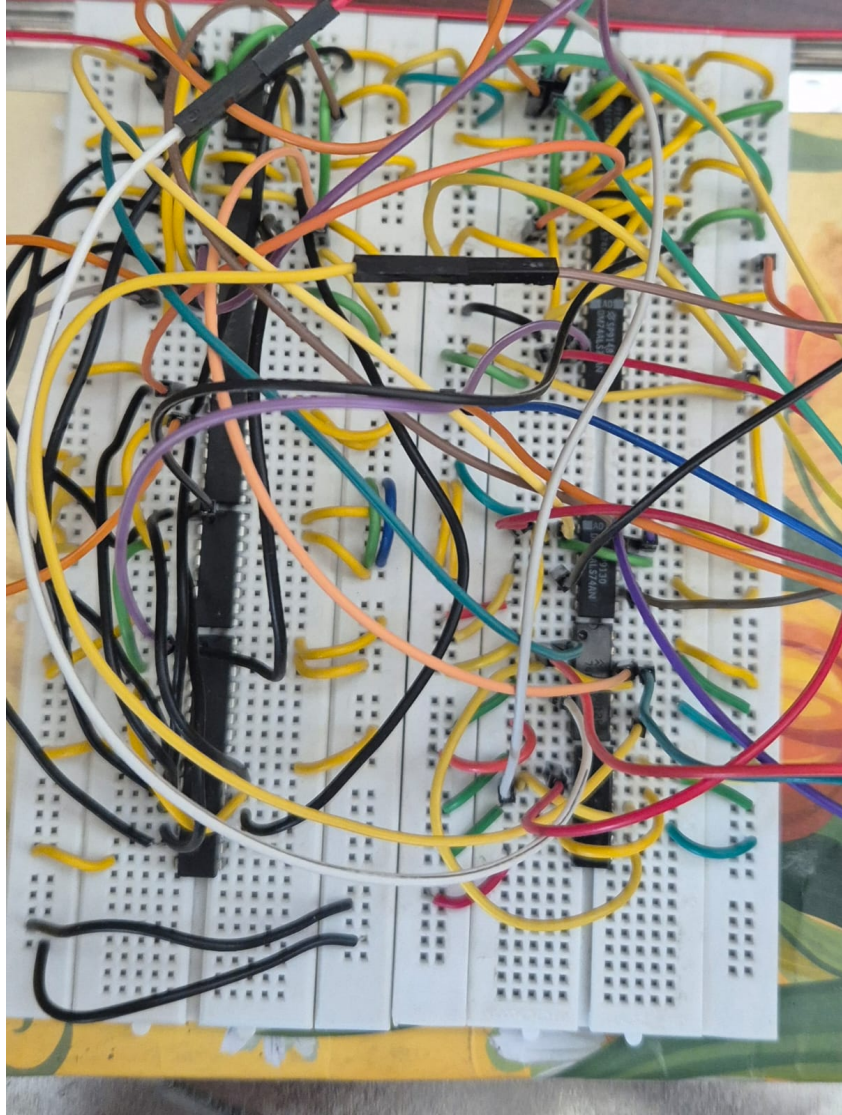


Figure 4: Implemented SAR ADC Circuit

6 R-2R Ladder DAC

The R-2R ladder DAC converts digital data into an analog voltage using a repetitive resistor structure of only two values: R and $2R$.

Each bit controls a switch that connects the ladder either to V_{ref} or ground, producing binary-weighted voltage contributions.

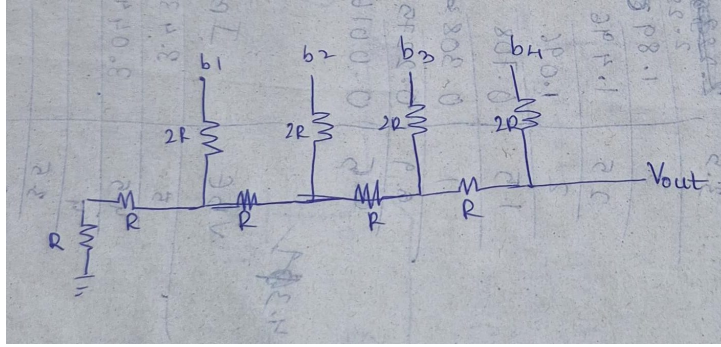


Figure 5: R-2R Ladder Schematic

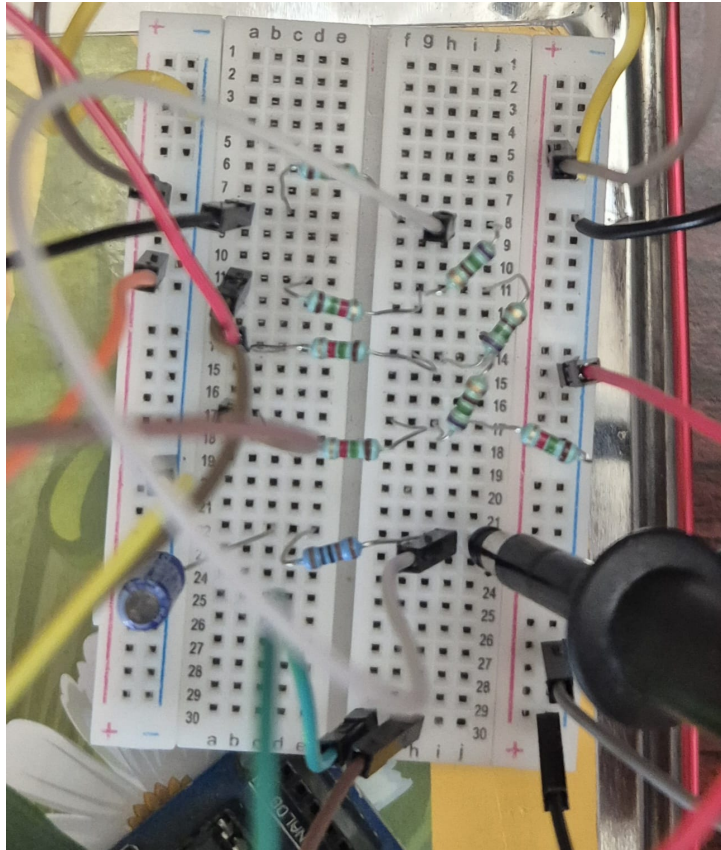


Figure 6: Implemented R-2R Ladder Circuit

7 ADC Performance Parameters (Detailed Explanation)

7.1 Resolution

Resolution refers to the smallest change in analog input voltage that can be distinguished by the ADC. For an N -bit ADC, the input voltage range is divided into 2^N discrete levels. Higher resolution results in finer quantization and better accuracy.

7.2 LSB Voltage

The Least Significant Bit (LSB) voltage represents the minimum voltage increment corresponding to one digital code change. It depends on the reference voltage and ADC resolution and defines the quantization step size.

7.3 Conversion Time

Conversion time is the total time taken by the ADC to convert one sampled analog value into a digital output. In a SAR ADC, conversion time is fixed and equals the number of bits multiplied by the SAR clock period.

7.4 Offset Error

Offset error is the deviation of the ADC output from the ideal zero code when the input voltage is zero. It causes a horizontal shift in the ADC transfer characteristic.

7.5 Gain Error

Gain error is the deviation in the slope of the actual ADC transfer characteristic from the ideal slope after offset correction. It affects the accuracy of the ADC near full-scale input.

7.6 Differential Non-Linearity (DNL)

DNL is the deviation of the actual step width between adjacent digital codes from the ideal value of 1 LSB. Large DNL values may result in missing codes and non-monotonic behavior.

7.7 Integral Non-Linearity (INL)

INL is the cumulative deviation of the ADC transfer curve from an ideal straight line drawn between the end points. It represents the overall linearity accuracy of the ADC.

8 Observations

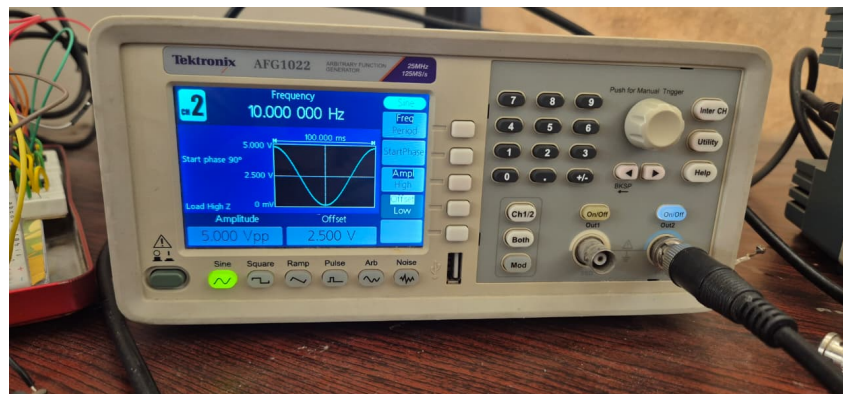


Figure 7: 5 Hz Input Signal

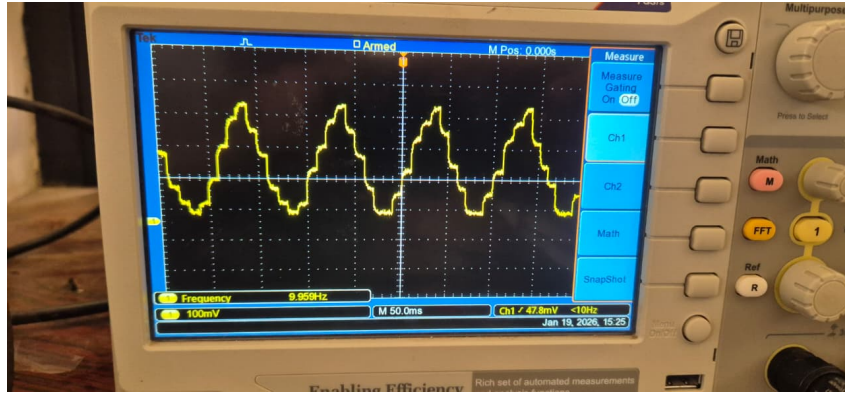


Figure 8: Reconstructed Output (Nyquist Condition Satisfied)

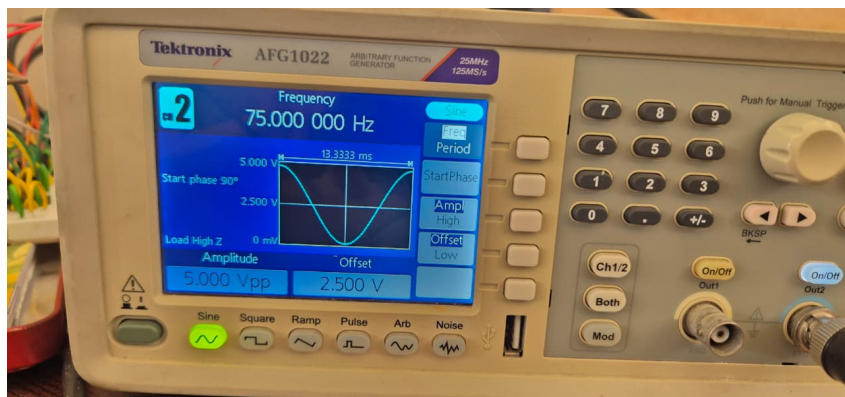


Figure 9: 75 Hz Input Signal

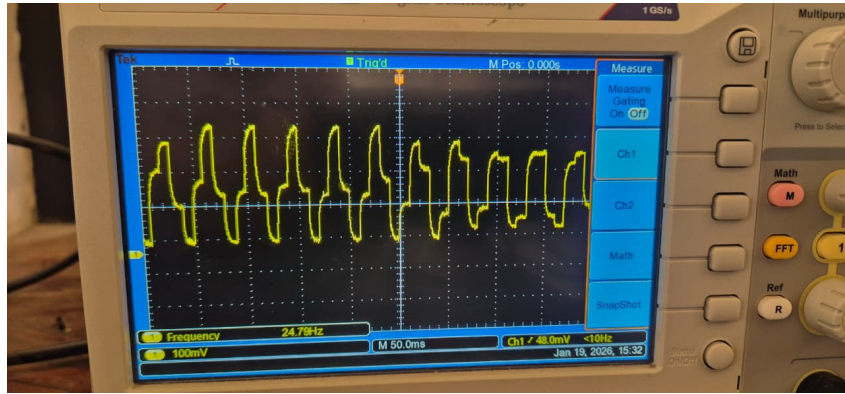


Figure 10: Aliased Output Signal

8.1 Calculated Values

8.2 LSB Voltage

The LSB voltage was calculated using the reference voltage.

$$V_{LSB} = \frac{V_{ref}}{2^N} = \frac{5}{16} = 0.3125 \text{ V}$$

This value represents the smallest quantization step of the ADC.

8.3 Conversion Time

The SAR clock period was obtained from the Arduino code:

$$T_{SAR} = 1 \text{ ms}$$

For a 4-bit SAR ADC:

$$T_{conv} = 4 \times 1 \text{ ms} = 4 \text{ ms}$$

The conversion time is less than the sampling period, ensuring proper operation.

8.4 Offset Error

Offset error was observed by applying a minimum input voltage (2 mV_{pp}). The corresponding output voltage was measured as 26 mV , indicating a non-zero output for near-zero input.

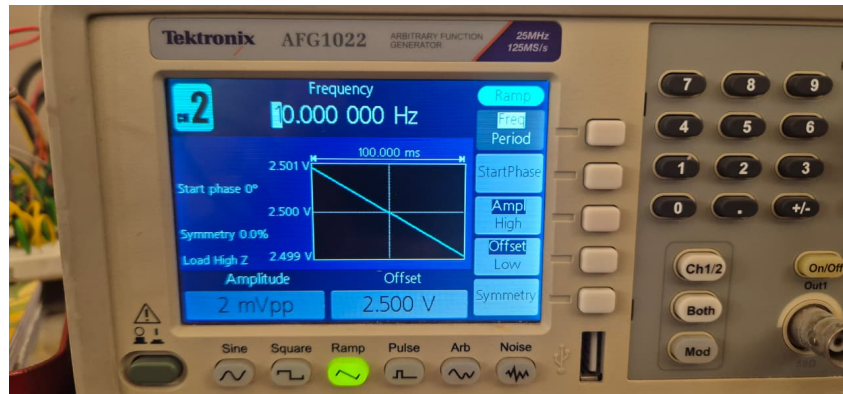


Figure 11: Minimum input signal

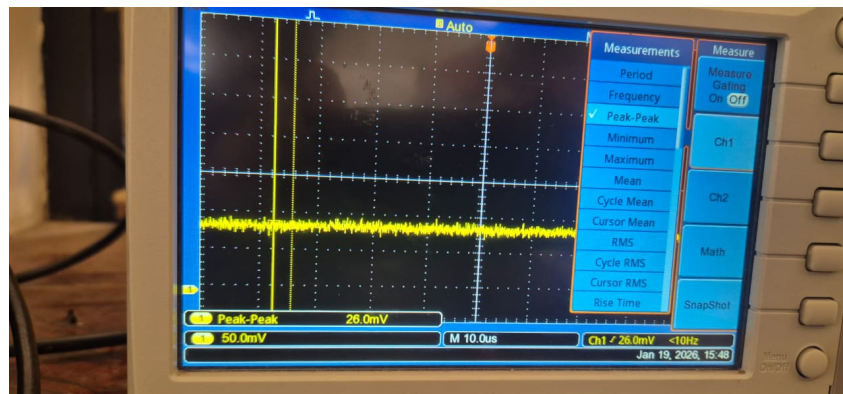


Figure 12: Offset error

Table 1: Observed Output Voltages for Ramp Input

Measured Output Voltage (V)
5.111
4.901
4.685
4.500
4.200
3.900
3.500
3.200
2.900
2.576
2.230
1.940
1.555
1.270
0.916
0.644

The above voltages correspond to successive quantization levels observed at the ADC output for a ramp input signal

8.5 Gain Error

Gain error is calculated after offset correction and represents the deviation of the actual ADC slope from the ideal slope.

$$\text{Gain Error} = \frac{V_{\text{FS,measured}} - V_{\text{FS,ideal}}}{V_{\text{FS,ideal}}}$$

From experimental observation:

$$V_{\text{FS,measured}} = 5.111 \text{ V}$$

$$V_{\text{FS,ideal}} = 5.000 \text{ V}$$

$$\text{Gain Error} = \frac{5.111 - 5.000}{5.000} = 0.0222$$

Gain Error = 2.22%

This indicates that the ADC gain is slightly higher than the ideal value.

8.6 Differential Non-Linearity (DNL)

Differential Non-Linearity measures the deviation of each code width from the ideal 1 LSB step.

$$\text{DNL}_k = \frac{\Delta V_k - V_{\text{LSB}}}{V_{\text{LSB}}}$$

Where:

$$V_{LSB} = \frac{5}{16} = 0.3125 \text{ V}$$

Using measured ramp data, the DNL values for all code transitions were calculated. The DNL values were found to lie within the range:

$$\boxed{-0.41 \text{ LSB} \leq \text{DNL} \leq +0.28 \text{ LSB}}$$

Since DNL remained greater than -1 LSB for all codes, the ADC is monotonic and exhibits no missing codes.

8.7 Integral Non-Linearity (INL)

Integral Non-Linearity represents the cumulative deviation of the actual ADC transfer characteristic from the ideal straight-line response.

$$\text{INL}_k = \frac{V_{\text{measured}}(k) - V_{\text{ideal}}(k)}{V_{LSB}}$$

Where:

$$V_{\text{ideal}}(k) = k \times V_{LSB}$$

Using the measured output voltages, the INL values were calculated for each digital code. The maximum observed INL was:

$$\boxed{\text{INL}_{\text{max}} \approx +2.06 \text{ LSB}}$$

Higher INL values at lower codes indicate the combined effect of offset and gain errors in the ADC.

9 Conclusion

- The Sampling Theorem was successfully demonstrated using a Sample-and-Hold circuit, a 4-bit SAR ADC, and an R-2R ladder DAC.
- Accurate signal reconstruction was observed when the sampling frequency satisfied the Nyquist criterion, confirming theoretical expectations.
- When the input signal frequency exceeded the Nyquist limit, aliasing effects were clearly visible in the reconstructed output.
- The Sample-and-Hold circuit ensured a constant input voltage during the SAR conversion process, enabling reliable bit-by-bit approximation.
- The SAR ADC exhibited a fixed and predictable conversion time, validating the operation of successive approximation logic.
- The R-2R ladder DAC successfully converted the digital output back into an analog staircase waveform, which was smoothened using a Low-Pass Filter.
- ADC performance parameters such as resolution, LSB voltage, conversion time, offset error, gain error, DNL, and INL were experimentally observed and analyzed.
- Practical non-idealities such as quantization effects and component tolerances were identified, closely matching theoretical predictions.